

Project: FIFA 23 Players

Applied Statistics and Data Analytics

April 25, 2026

1 Background description

FIFA is a popular football game which collects information about football (soccer) players from around the world and measures players' abilities with explicit numbers. The dataset below offers a wealth of attributes related to each player, making it a valuable resource for various analyses and insights into the realm of football, both for gaming enthusiasts and real-world sports enthusiasts.

In this dataset, you can study the relationship among different attributes. The attribute "overall" and "potential" seem to be comprehensive descriptions of a player's ability, so you can try to study how FIFA estimates the value of a player by fitting models like a regression model. Moreover, the dataset includes each player's position and best position. You can find out the relationship between player's abilities and positions.

2 Data description

FIFA 23 Players.csv : The data is collected from kaggle. It contains 17524 different records with 89 attributes.

Details of some attributes:

- Known As: The nickname or commonly used name of the player.
- Full Name: The complete name of the player.
- Overall: The player's overall rating out of 100, based on their performance in different aspects of the game.
- Potential: The player's potential rating out of 100, indicating how much their overall rating can improve with training and experience.
- Value (in Euro): The estimated monetary value of the player in euros, based on their age, skill level, and market demand.
- Positions Played: The different positions on the field where the player can play.
- Best Position: The position where the player performs best, based on their skills and abilities.

- Nationality: The country the player represents in international soccer matches.
- Image Link: The link to the player's image.
- Age: The player's age in years.
- Height (in cm): The player's height in centimeters.
- Weight (in kg): The player's weight in kilograms.
- TotalStats: The sum of all the player's statistics, which include physical attributes, skills, and abilities.
- BaseStats: The sum of the player's fundamental attributes, such as speed, shooting, passing, dribbling, and defending.
- Club Name: The name of the player's current club.
- Wage (in Euro): The player's weekly salary in euros.
- Release Clause: The amount of money required to buy the player's contract from their current club.
- Club Position: The player's position in their current club.
- Contract Until: The year when the player's contract with their current club expires.
- Club Jersey Number: The number on the player's jersey.
- Joined On: The year when the player joined their current club.
- On Loan: Indicates whether the player is currently on loan to another club.
- Preferred Foot: The foot the player uses most often to kick the ball.
- Weak Foot Rating: A rating out of five indicating the player's proficiency with their weaker foot.
- Skill Moves: A rating out of five indicating the player's proficiency with advanced dribbling and ball control techniques.
- International Reputation
- National Team name: The national team the player belongs to.
- National Team Image Link
- National Team Position: The position where the player performs in the national team.
- National Team Jersey Number: The jersey number in the national team.
- Attacking Work Rate: The tendency to join in attack work.
- Defensive Work Rate: The tendency to join in defense work.

There are 6 attributes describing a player's ability comprehensively: "Pace Total", "Shooting Total", "Passing Total", "Dribbling Total", "Defending Total" and "Physicality Total". In the file "*Fifa_23_Players_Data.csv*", The attributes from "Crossing" to "Sliding Tackle" (29 attributes in total) are used to describe a player's ability in detail.

If a player plays as GK (Goalkeeper), only 7 attributes will be used: "Acceleration", "Sprint Speed", "Goalkeeper Diving", "Goalkeeper Handling", "Goalkeeper Kicking", "Goalkeeper Reflexes" and "Goalkeeper Positioning". The attributes from "ST Rating" to "GK Rating" (17 attributes in total) describe the ratings of the player if he or she plays in a specific position.